Basic Principles of Medicine 1 Module: Foundations to Medicine Lecture 27

Lecture Title: Population Genetics

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Pre-lecture Recommended Reading

- **Human Genetics and Genomics, Korf and Irons, 4th Edition Chapter 7**
  - **You can find the chapter on your Sakia site under:**
  - Resources/General Information/Book Access

Post-lecture DLA Objectives

- DLA 16 linked to this lecture

SOM.MK.III.BPM1.1.FTM.3.GNET.0086	Analyze how alleles segregate in populations and how to relate allele frequency in populations to genotype frequency of the people in the population
SOM.MK.III.BPM1.1.FTM.3.GNET.0087	Apply the Hardy Weinberg equation to clinical scenarios to allow estimation of carrier risk, recurrence risk, and frequency of disease alleles in a population for AR, AD, and X-linked genes.
SOM.MK.III.BPM1.1.FTM.3.GNET.0088	Estimate a carrier frequency from the Hardy Weinberg equation and population incidence for one individual to estimate recurrance risk when that person mates with an individual who has familial carrier risk.
SOM.MK.III.BPM1.1.FTM.3.GNET.0089	Use Hardy-Weinberg equation to demonstrate why the expected frequency of rare autosomal dominant alleles are expected to be extremely infrequent.

Email: mmaj@sgu.edu (Questions regarding lecture and DLA)

Objectives

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SOM.MKIII.BPM1.1.FTM.3.GNET.0089	Use Hardy-Weinberg equation to demonstrate why the expected frequency of rare autosomal dominant alleles are expected to be extremely infrequent.

Population Genetics is like a "null model for evolution"

Hardy-Weinberg Equilibrium (HWE),

Stable, non-evolving population remains the same and you can predict the frequency alleles in the population

DLA discusses deviations from the HWE and will be tested on in exams and iMCQs

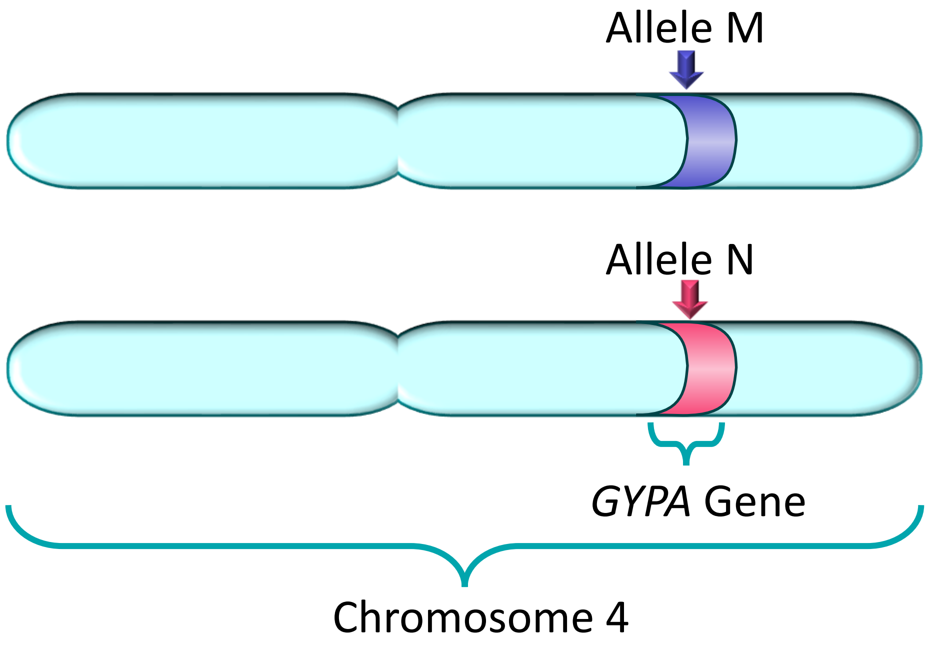
Definitions

- **Population genetics:** Study of genetic variation within populations and evolutionary factors that explain this variation
    - Hardy-Weinberg law – population is large, and alleles will be in equilibrium if mating is random and no mutation, no selection nor migration occurs
- **Alleles:** A variant of the same gene, most often a substitution of a base
- **Polymorphism:** Multiple forms (alleles) of a gene in a population, **regular occurrence**, usually > 1% of the population, usually a single nucleotide change, can be used as molecular markers
- **Pathogenic Mutations:** usually a single nucleotide change, **rare**, < 1% of the population
- **Homozygote:** Same allele on each chromosome at a specific locus
- **Heterozygote:** Different alleles of a specific gene at a specific locus
- **Epidemiological Studies:** Determination of a specific disease in a defined population which will identify the **Incidence** $(\boldsymbol{I})$ of a specific disease in that specific population
- **Incidence** $(\boldsymbol{I})$ is the number of people in a specific population with a specific disorder

For the purposes of this lecture, the terms "male" and "female" are narrowly defined as the individual's biological sex at birth as it determines clinical care

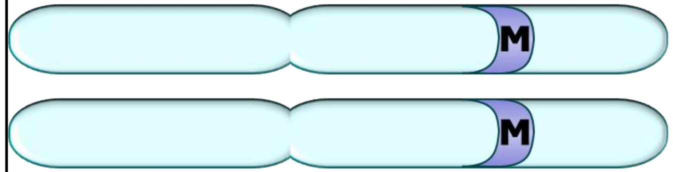
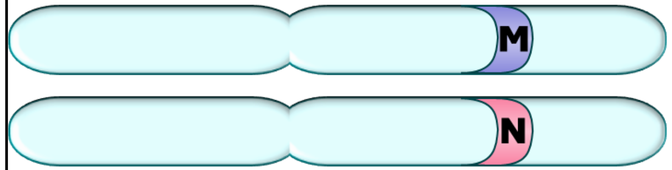
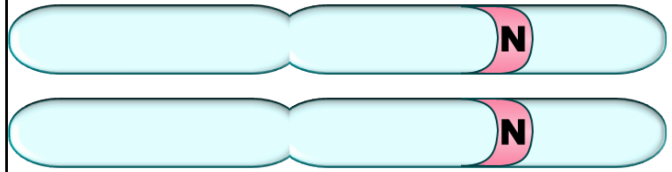
Genetic Variation in Populations

- DNA sequence differences in a gene may result in different forms of a protein, but not all differences affect an amino acid
- Most commonly, autosomal trait is regulated by single pair of variants on one gene
- The Frequency of the two alleles combine into genotypes randomly
  - Example of a gene with **polymorphic** *GYPA* gene is the MN blood group (Co-dominant)



3 single nucleotide polymorphism (SNP) variations in the gene sequence leads to 2 amino acid changes

Genotypes in a Population of 100 People:

	Homozygous for Allele M Genotype MM	64 people were found to be homozygous for M
	Heterozygous Genotype MN	32 people were found to be heterozygous
	Homozygous for Allele N Genotype NN	4 people were found to be homozygous for N

Allele Frequency from Same Data
Recall We Are Biallelic = 200 alleles for 100 people

Genotype	Number of People	Genotype frequency	# of alleles	Add up the alleles	Allele Frequency
MM	64	0.64 (64/100)	**64 = M** **64 = M**	160 alleles are M	Frequency of **M (p)** 160/200 = 0.8 or 80%
MN	32	0.32 (32/100)	**32 = M** **32 = N**		
NN	4	0.04 (4/100)	**4 = N** **4 = N**	40 alleles are N	Frequency of **N (q)** 40/200 = 0.2 or 20%
Total	100	1.00		200 Alleles	1.00

Allele Frequency

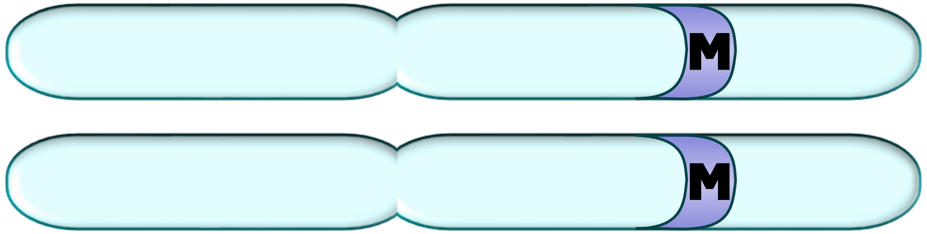
- Two allele system: a given population has 2 alleles:
  - Previous example used alleles **M** and **N**
  - Patterns of inheritance uses **A** and **a** alleles
  - Genotype calculation uses **p** and **q** alleles
- The sum of gene frequencies, all cases, must be 1

$$\mathbf{p + q = 1}$$

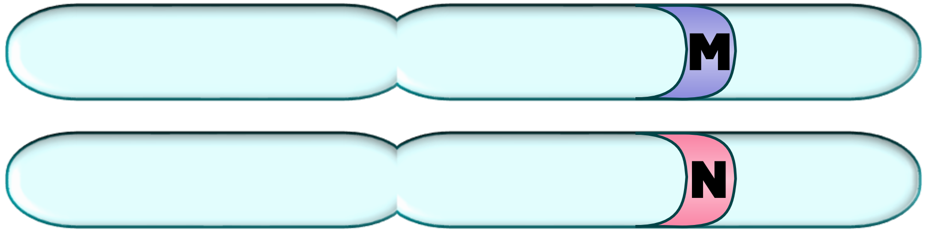
- From previous example, allele frequency of **p** = 0.8
- Therefore, the frequency of **q** = 0.2 according to the equation

$$\boxed{\mathbf{0.8 + 0.2 = 1}}$$

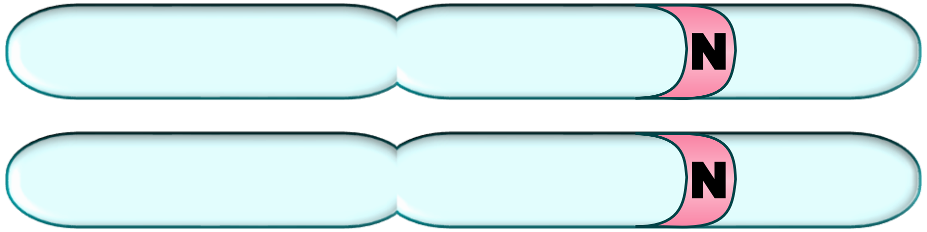
Recall Genotypes Frequency in the Population from previous slide



Homozygous for Allele M
Genotype MM = 64 people



Heterozygous
Genotype MN = 32 people



Homozygous for Allele N
Genotype NN = 4 people

Hardy-Weinberg Equilibrium

To Determine Genotype Frequency from Incidence ($\boldsymbol{I}$)

Assumptions of the Equilibrium, allele frequencies do not change over time provided:

Large population (no genetic drift),

Random mating,

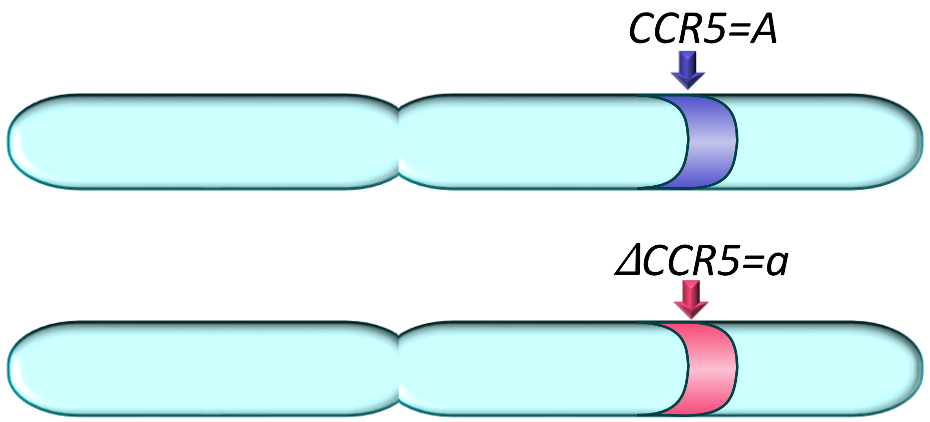
No selection (phenotypes have equal fitness),

No migration

No mutation

Allele vs. Genotype Frequencies

- Most commonly, autosomal trait regulated by a single pair of alleles for one gene
- Example: Resistance to HIV by deletion of 32 bp in the *CCR5* gene = *ΔCCR5*



Genotype	# People	Genotype Frequency	Add up Alleles
CCR5/CCR5 or AA	647	0.82	CCR5 alleles = 1294
CCR5/ΔCCR5 or Aa	134	0.17	CCR5 alleles = 134 *ΔCCR5 alleles = 134*
ΔCCR5/ΔCCR5 or aa	7	0.01	*ΔCCR5 alleles = 14*
Total	788	1	(788 x 2) = 1576

Ratios

CCR5 allele (A) = common allele = $^{1428}/_{1576} = 0.91$

ΔCCR5 allele (a) = rare allele = $^{148}/_{1576} = 0.09$

Hardy-Weinberg:
Calculate Genotype Frequency from Allele Frequency using p & q

The chance that 2 A alleles will pair up to give AA genotype is p^2
The chance that 2 *a* alleles will pair up to give *aa* genotype is q^2

Hardy-Weinberg states the frequency of three genotypes: **AA**, **Aa** and **aa** is given by the terms of the binomial expansion $(p + q)^2 =$ $\mathbf{pp + pq + qp + qq = 1}$

Simplified to the HW Equation $\mathbf{= p^2 + 2pq + q^2 = 1}$

If → ***CCR5*** **= p = 0.91 and** ***ΔCCR5*** **= q = 0.09**

Then → $\mathbf{0.91^2 + 2(0.91 \times 0.09) + 0.09^2 = 1}$

And → **0.828 + 0.164 + 0.008 = 1**

Let "p" be the functional or common allele
Let "q" be the disease causing or rare allele

Can you calculate Allele Frequency or Predict Genotype Frequency if you Calculate the percentage of Affected People in a Population?

Yes, by applying Hardy-Weinberg Equation to a Population in Equilibrium

And knowing the Pattern of Inheritance

We will now be discussing rare mutations, so the allele is found in less than 1% of the population and greater than 99% of the population do not have the mutation
Therefore, the normal allele is about 100%

Genetic Disorders are Rare

- Example for autosomal recessive disorder
- The allele frequency for medium-chain acyl-CoA dehydrogenase deficiency (MCAD) is 0.007 = q (rare disease allele)

$$p + q = 1$$

$$p = 1 – 0.007$$

$$p = 0.993$$

p is the common allele
Which is really close to 1

To simplify most of our calculations in this lecture, we make the common allele = 1 so we can drop it out of our multiplication steps

Application of HW: Autosomal Recessive Disorders

- Autosomal Recessive:
  - p^2 are homozygous normal persons
  - $2pq$ are heterozygotes, or carriers, no apparent disease persons
  - q^2 is homozygous affected persons

$$\boxed{p^2 + 2pq + q^2 = 1}$$

Affected Persons in the population from epidemiological studies

"Incidence" of the disorder

AR: Solving for 2pq when you have Incidence

If $q^2 = I$,

then solve for q $\boxed{q = \sqrt{I}}$

From q, you can calculate 2pq (carriers)

Must assume that q^2 (affected persons) are rare in the population $\leq 1\%$

and thus, **p is ≈ 1**

AR: Example PKU has incidence of 1/10,000

$q = \sqrt{I} = \sqrt{{}^{1}/_{10,000}} = {}^{1}/_{100}$

Calculating 2pq when we know PKU $\leq$ 1%

$\therefore p \cong 1$ $\boxed{\mathbf{p^2 + 2\cancel{p}q + q^2 = 1}}$

And $2(q) = 2\,({}^{1}/_{100}) = {}^{2}/_{100} = {}^{1}/_{50}$

: If $\boldsymbol{I}$ of PKU = ${}^{1}/_{10,000}$ then ${}^{1}/_{50}$ people are carriers in that population

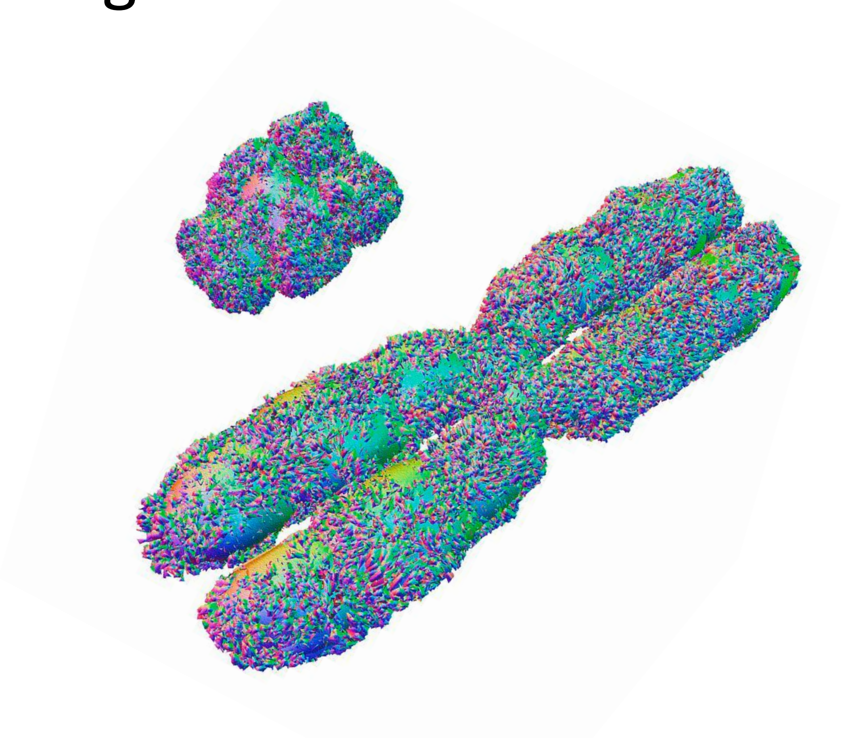
Application of HW: X-linked Disorders

For the purposes of this lecture, the terms "male" and "female" are narrowly defined as the individual's biological sex at birth as it determines clinical care

These disorders are mainly expressed in **live born males:**

Males are heterogametic for X-chromosome, "hemizygous"

Females are homogametic for X-chromosome



~290 X-linked recessive conditions
~10 X-linked dominant conditions

Things that confound definition of X-linked dominant for females: skewed X-inactivation & reduced penetrance

Many exceptions for X-linked disorders

Application of HW: X-linked

- Males are hemizygous for X-linked genes
- Referring to the HW equation, it can be applied to X-linked disorders
  - p^2 are homozygous normal females
  - 2pq are female heterozygotes, or carriers, no apparent disease (mostly)
  - q is affected males

$$p^2 + 2pq + q^{\cancel{2}} = 1$$

Affected Males in the population from epidemiological studies

"Incidence" of the disorder

X-linked: Solving for 2pq when you have Incidence

If q = I,

then… well…

solve for q $\boxed{q = I}$

From q, you can calculate 2pq

Must assume that q (affected persons) is rare in the population

and thus, **p is ≈ 1**

X-linked: Example Hemophilia has incidence of 1/10,000

$q = I = {}^{1}/_{10,000}$

Calculating 2pq (female carriers) when we know Hemophilia $\leq$ 1%

$\therefore p \cong 1$

$$\boxed{p^2 + 2\cancel{p}q + q^2 = 1}$$

And $2q = 2\,({}^{1}/_{10,000}) = {}^{2}/_{10,000} = {}^{1}/_{5000}$

: If I of Hemophilia = ${}^{1}/_{10,000}$ then ${}^{1}/_{5000}$ female carriers in that population

Application of HW: Autosomal Dominant Disorders

- Autosomal Dominant:
  - p^2 are homozygous normal persons
  - 2pq are heterozygotes, disease persons
  - q^2 is homozygous affected persons (not usually compatible with life but includes FHC)

$$p^2 + 2pq + \cancel{q^2} = 1$$

Affected Persons in the population from epidemiological studies

"Incidence" of the disorder

AD: Solving for q^2 when you have Incidence

If $2pq = I$,

then solve for q $\boxed{q = {}^{1}\!/_{2} \times I}$

From q, you can calculate q^2

Must assume that q^2 (affected persons) are rare in the population

and thus, **p is ≈ 1**

AD: Example Huntington has incidence of 1/10,000

q = $^{1}/_{2}\, x\, I$ = $^{1}/_{2}\, x\, ^{1}/_{10,000}$ = $^{1}/_{20,000}$

Calculating q^2 when we know HD $\leq$ 1%

$\therefore$ p $\cong$ 1 $\boxed{p^2 + 2\cancel{p}q + q^2 = 1}$

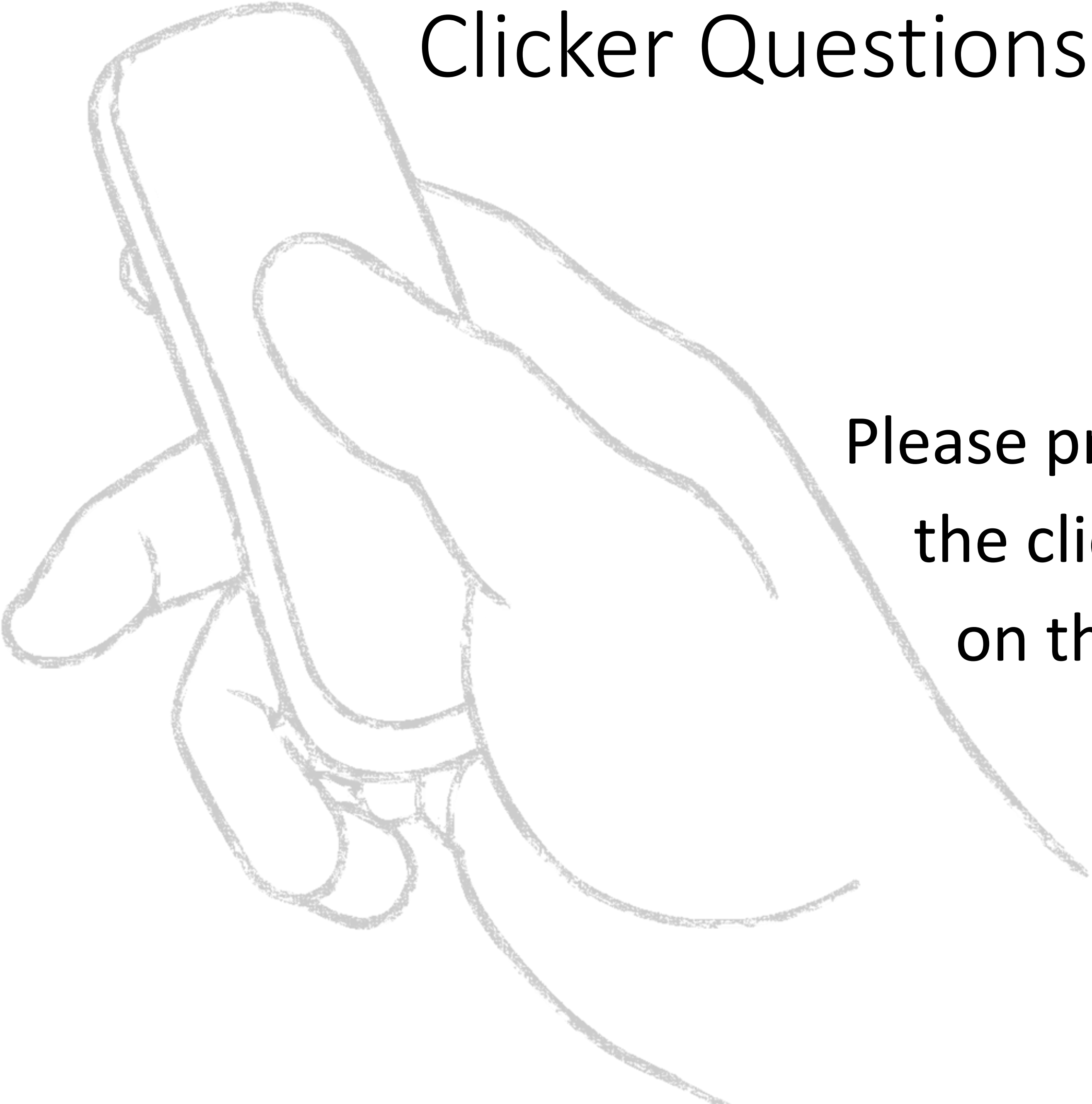
And $q^2 = (^{1}/_{20,000})^2 = {}^{1}/_{4 \text{ x } 10^8}$

: If I of HD = $^{1}/_{10,000}$ then $^{1}/_{4\, x\, 10^8}$ people are homozygous in that population

Population Genetics Hardy-Weinberg Summary

Inheritance	Solving for q from Incidence	Using q to calculate other parameters
Autosomal Recessive	$\boldsymbol{q = \sqrt{I}}$	Carrier = 2q
X-Linked	$\boldsymbol{q = I}$	Female Carrier = 2q Homozygous Female = q^2
Autosomal Dominant	$\boldsymbol{q = {}^{1}/_{2} * I}$	Homozygous (Rare) = q^2

Clicker Questions Next!



Please prepare to answer
the clicker questions
on the next slides

Autosomal dominant disorders – **Homozygosity is Rare**

Homozygosity for AD alleles is very (exceedingly) rare and in almost all cases, the phenotype of the homozygote will be more severe the phenotype of the heterozygote.

- LDL-Receptor defects leading to FHC (*LDLR*)
  - Heterozygote (atherosclerosis in adulthood)
  - **Homozygote (much more severe and childhood onset)**
- Achondroplasia (*FGFR3*)
  - Heterozygote has short stature
  - **Homozygote has embryonic, fetal, or perinatal lethality**
- Huntington disease (CAG triplet repeat expansion in *HD* gene)
  - **In the few documented examples, homozygotes have earlier and more severe disease onset (Early childhood onset)**

Autosomal Recessive

- $q^2 = 1/2500$
  - Number of affected homozygote patients
  - Counted by disease incidence
- $q = \sqrt{^1/_{2500}}$ ~ = 1/50 (mutant allele frequency in population)
  - Note that 1/50 is a small number; (0.02, sometimes written as 2%)
  - p = 1- 1/50 = 49/50 (≈1)
- 2q = 2x (1/50) ~ = 1/25 : Frequency of heterozygous carriers in the population
  - thus about 4% of the Caucasian population is a carrier for CF

Solve this later on your own if we don't have time in lecture

Consider the following three disorders:

1. Autosomal Dominant Retinoblastoma
2. Autosomal Recessive Friedreich ataxia (a neuromuscular disorder)
3. X-linked choroideremia (a cause of vision loss in males)

Each has disease incidence of approximately 1/25,000

What is the allele frequency and heterozygote frequency for each?

Answers

Inheritance	Solving for q from Incidence	Using q to calculate other parameters
Autosomal Recessive	$q = \sqrt{I}$	Carrier = 2q
X-Linked	$q = I$	Female Carrier = 2q Homozygous Female = q^2
Autosomal Dominant	$q = {}^{1}/_{2} * I$	Homozygous (Rare) = q^2

Retinoblastoma (AD)

- $2pq = 1/25{,}000$, $p \sim 1$
- Thus $q = 1/50{,}000$;
- $q^2 = {}^{1}/_{2.5\times10^{8}}$

Friedreich ataxia (AR)

- $q^2 = 1/25{,}000$
- $q = 1/158$, $p \sim 1$,
- $2pq = 1/79$

Choroideremia (X-linked)

- $q = 1/25{,}000$
- And of course, female carriers would be → $2pq = 1/12{,}500$

The End